WashU McKelvey Engineering

Spring 2026 MEMS 3420
Heat Transfer Final Project

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Group 13

We hereby certify that the lab report herein was completed in accordance with the McKelvey School of Engineering and Undergraduate Student academic integrity policies. We attest that artificial intelligence (AI) tools were used in generating MATLAB code for data visualization and refining the formatting and wording of the Overleaf report. All engineering analysis, COMSOL simulations, calculations, and technical interpretations were completed independently by the authors. AI-generated content was reviewed and edited to ensure accuracy and consistency with the project results.

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Introduction

In this project, a heat sink, commonly used to cool down computer chips, will be analysed through COMSOL. By restricting the geometry to a rectangular base with a set amount of rectangular fins, all of identical size and evenly distributed, the project will explore various heat sink materials as well as different cooling fluids. All other convection flow parameters will remain constant, with a set fluid velocity that will guarantee laminar flow conditions for all tested fluids. The objective of these variations is to find the combination of materials (fin and fluid) that will generate the largest heat flux when placed on top of the same constant heat generation material, simulating a CPU chip. Thermal throttling is a prominent challenge in the design of electronics, and a proper analysis of what materials accomplish cooling best is essential for the use of modern processors that use a lot of power and produce a lot of heat. Examining the tested materials will provide insight and analysis into what material properties produce the best materials for heat transfer under forced convection constraints.

Pre-Analysis

In order to benchmark the COMSOL results, an analytical solution was generated for a 24-fin heat sink, under laminar flow conditions, assuming a conductive tip boundary condition for the individual fins. It was assumed that the heat rate going into the fin from its base (due to the CPU) is 90 W. Equation 1 describes the heat transfer rate of a fin with a convective tip boundary condition.

$$q_f = M\frac{sinh(mH) + (h/mk)cosh(mH)}{cosh(mH) + (h/mk)sinh(mH)} \tag{1}$$

q_f is the heat transfer rate of a singular fin [W], H is the height of the fin [m], m and M are defined by Eq. 2 and Eq. 3 respectively.

$$m = \sqrt{\frac{hP}{kA_c}} \tag{2}$$

where h is the convective heat transfer coefficient [W/m^{2}K], P is the perimeter of the cross section [m], k is the thermal conductivity of the fin [W/m·k], and A_c is the cross sectional area of the fin [m^2].

$$M = \sqrt{hPkA_c}(T_b - T_\infty) \tag{3}$$

T_b is the temperature of the base of the fin, and T_∞ is the temperature of the surrounding fluid.

To simplify the problem, evaluate the average Nusselt Number according to Eq. 4 along the length of the base plate.

$$\overline{Nu_L} = 0.664Re_L^{1/2}Pr^{1/3} \tag{4}$$

Re_x is the Reynolds Number and Pr is the Prandtl Number, defined by Eq. 5 and Eq. 6 respectively.

$$Re_L = \frac{u_\infty L}{\nu} \tag{5}$$

u_∞ is the incident velocity [m/s], L is the length of the base plate [m] and ν is kinematic viscosity [m^2/s].

$$Pr = \frac{\nu}{\alpha} \tag{6}$$

α is thermal diffusivity [m^2/s]. Solving for $\overline{h}$, utilize the definition of a Nusselt Number Eq. 7.

$$\overline{Nu_L} \equiv \frac{\overline{h}L}{k_f} \tag{7}$$

k_f is the thermal conductivity of the fluid [W/m·K]. Equation 8 is an energy balance for the heat sink system, which will be used to solve for T_b.

$$q_{tot} = 90\ \text{W} = q_{base} + 24 * q_f = hA_{exp}(T_b - T_\infty) + \sqrt{hPkA_c}\frac{sinh(mH) + (h/mk)cosh(mH)}{cosh(mH) + (h/mk)sinh(mH)}(T_b - T_\infty) \tag{8}$$

A_{exp} is the area of the bottom plate that is not covered by the fins [m^2].

Equation 9, Eq. 10, and Eq. 11, can be used to evaluate the geometric components of the problem P, A_c, and A_{exp} respectively.

$$P = 2(w + t) \tag{9}$$

w is the width of one fin [m], t is the thickness of a fin [m].

$$A_c = w * t \tag{10}$$

$$A_{exp} = L^2 - 24 * A_c \tag{11}$$

The overall fin efficiency, η_o can be evaluated according to Eq. 12

$$\eta_o = \frac{q_{tot}}{q_{max}} = \frac{q_{tot}}{\overline{h}A_{tot}(T_b - T_\infty)} \tag{12}$$

where A_{tot} is the total area of the plate and the fin surface areas [m^2], defined by Eq. 13.

$$A_{tot} = L^2 + 24 * P * H \tag{13}$$

Table 1 summarizes the known numerical parameters of the problem. Properties were evaluated at an assumed film temperature of 350 K [1].

Table 1 Analytical solution numerical parameters

Parameter	Symbol	Units	Value
Thermal Conductivity	k, k_f	W/m·K	Air: 0.03, Water: 0.668, 6601 Aluminum: 162, Copper: 407
Incident Velocity	u_∞	m/s	2.00
CPU Heat Transfer Rate	q_{tot}	W	90.0
Air Properties			
Kinematic Viscosity	ν	m^2/s	$20.92*10^-6$
Thermal Diffusivity	α	m^2/s	$29.9*10^-6$
Water Properties			
Kinematic Viscosity	ν	m^2/s	3.789*10^-7
Thermal Diffusivity	α	m^2/s	1.635*10^-7
Geometric Properties			
Length of Base Plate	L	mm	60.0
Fin Width	w	mm	4.00
Fin Thickness	t	mm	1.00
Fin Height	H	mm	40.0

Table 2 summarizes the heat sink change in temperature, analytical efficiency, and fin heat transfer rate results for the two materials.

Table 2 Analytical Results

Heat Sink Material	Convective Heat Transfer Coefficent, h [W/m^2K]	Temperature Difference, (T_b - T_∞) [K]	Fin Heat Transfer Rate, q_f [W]	Overall Fin Efficiency, η_o
Air Results				
6601 Aluminum	22.32	344.1	2.628	0.8875
Copper	22.32	321.6	2.702	0.9497
Water Results				
6601 Aluminum	5505	3.778	0.7135	0.3278
Copper	5505	3.399	1.018	0.3643

Considering these analytical results, it is expected that the copper heat sink will be more efficient than the aluminum heat sink, as copper had analytical efficiencies of 95.0% (air) and 36.4% (water), and aluminum had 88.8% (air) and 32.8% (water) respectively. It is also expected that the heat sink will reach very high temperatures under these conditions for air, as the ΔT for the heat sinks were 344 K for 6601 aluminum and 322 K for copper, which is highly undesirable in real life. For those same conditions with water as the cooling fluid, however, the ΔT values were 3.78 K for 6601 Aluminum, and 3.40 K for copper which is far more reasonable for real-life applications. This is consequent of the

far larger convective heat transfer coefficient water has under these conditions, which is 5505 W/m^{2}K compared to 22.32 W/m^{2}K for air. Therefore, it is expected that the COMSOL results will similarly prove that water is the preferred cooling fluid, and copper is the preferred fin material.

Geometry

Figure 1 shows the geometry used in our model.

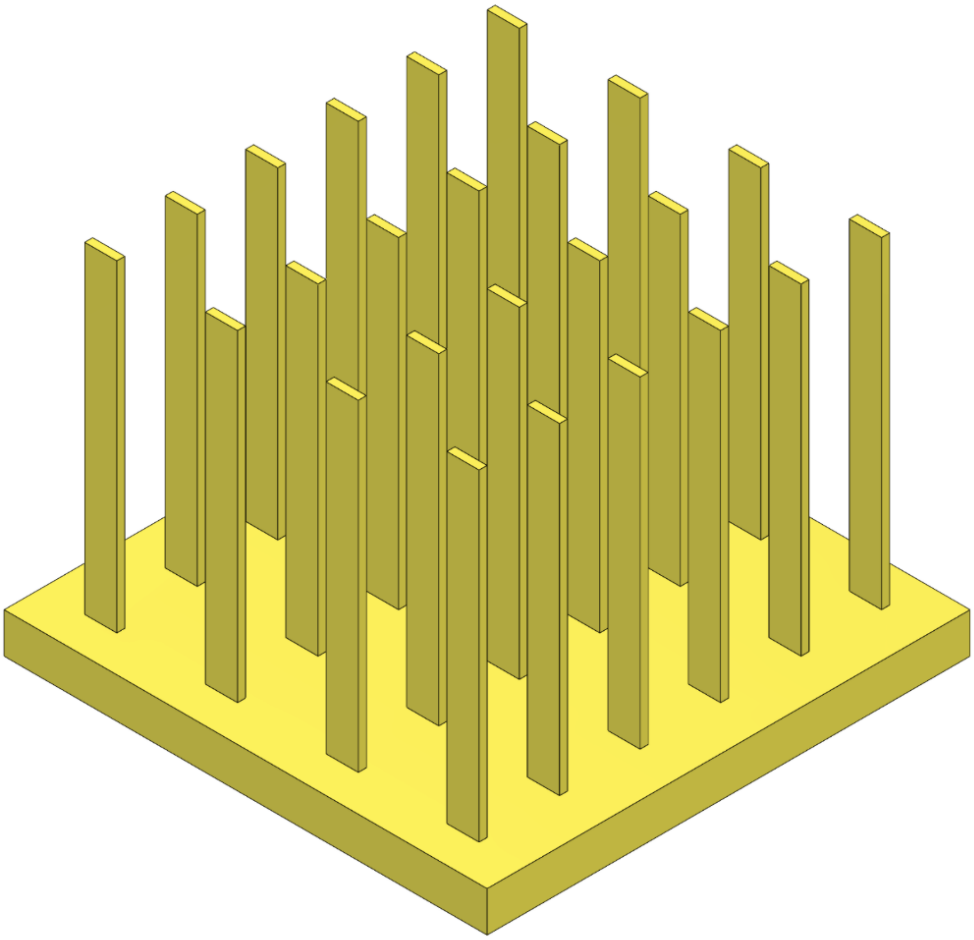


Figure 1 Heat sink geometry

Using SOLIDWORKS, the heat sink was modeled with 24 fins total, measuring 4 mm x 1 mm x 40 mm. The base plate measures 60 mm x 60 mm x 5 mm to be consistent with research. The fin spacing was determined by averaging out the distances of the fins between each other, ensuring a balanced layout that would not affect the results. Originally, a line of symmetry was going to be used to model only a quarter of the heat sink for simulation efficiency. However, many errors were encountered,

Mesh Level	Approx. Elements	Maximum Temperature [K]	Average h [W/m2K]
Coarser	45,000	NaN	NaN
Normal	250,000	294.84	63,004
Finer	750,000	294.82	63,085

Table 3 Effects of mesh size on temperature and convective heat transfer results

resulting in the whole geometry being imported into COMSOL to avoid any setbacks. A rectangular box was modeled in COMSOL around the heat sink in order to represent the working fluids, water and air.

Mesh

Figure 2 shows the fine mesh used in the COMSOL model for the heat sink.

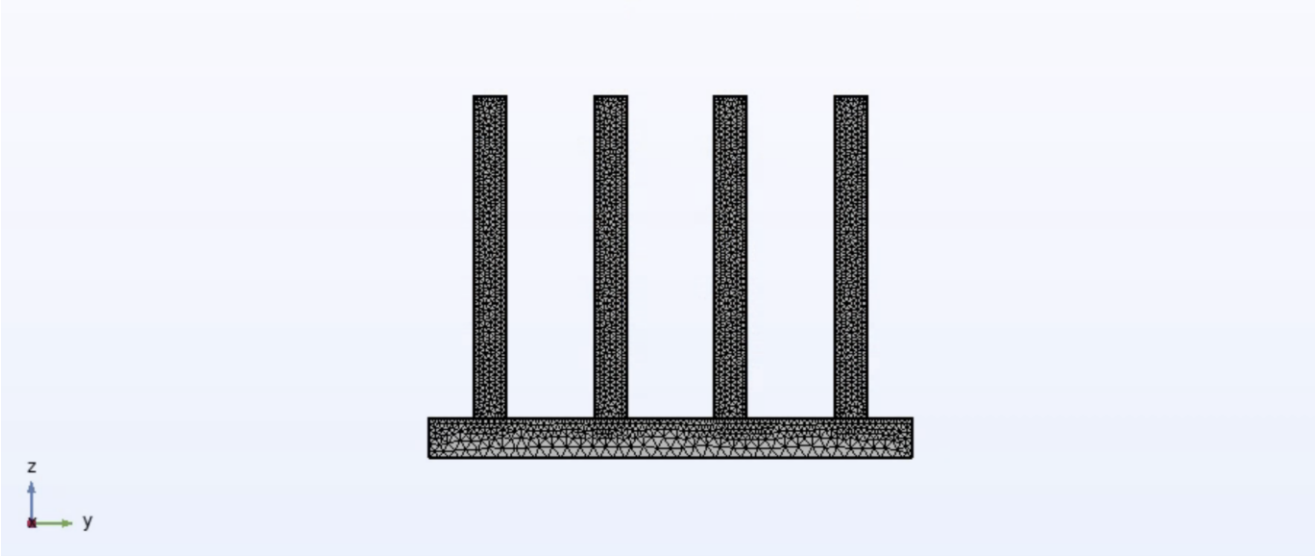


Figure 2 Heat sink in COMSOL with a fine mesh and balanced edges

Figure 3 shows the coarse mesh used in the COMSOL model for the working fluid boundaries. A tetrahedral mesh with normal setting was used for reasonable accuracy while not sacrificing the efficiency of the simulation. The element density was altered from the default normal setting at solid-fluid boundaries to accurately represent the hydrodynamic and thermal boundary layers. Additionally, the mesh of the heat sink was modeled with a fine mesh, while the air boundaries were modeled with a coarse mesh because of the emphasis on the results of the heat sink and to cut down simulation time. A mesh independence study was done to validate the mesh design. Coarser mesh configurations failed to converge due to unresolved gradients at boundary edges. The chosen mesh was correctly chosen due to a minimal temperature difference in comparison to a finer mesh study. Table3 displays values taken from a copper heat sink cooled by water.

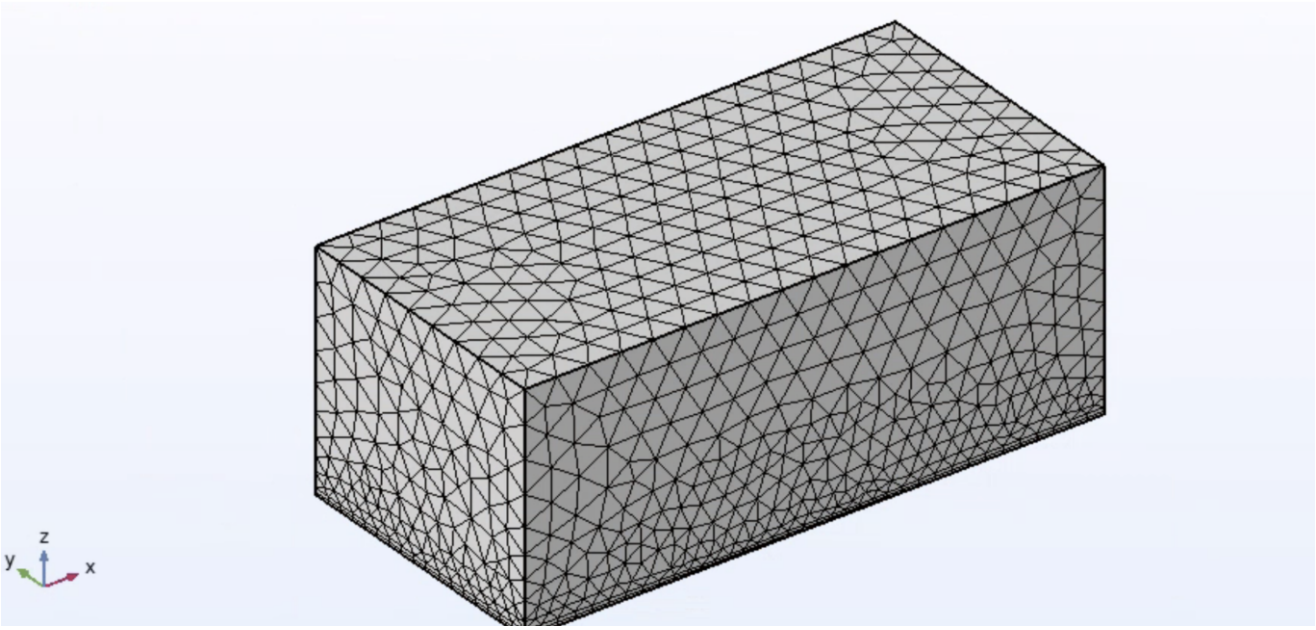


Figure 3 Working fluid boundaries with a coarse mesh and balanced edges

Setup

When setting up the COMSOL simulation, a Laminar Flow analysis and a Heat Transfer in Solids and Liquids analysis were chosen to find the effectiveness of the heat sink material and working fluids. Moreover, to connect the fluid and thermal aspects, a Non-Isothermal Flow multiphysics coupling was used to ensure that the fluid properties were allowed to change when the temperature varies. To represent a high thermal load of a CPU, a heat source of 90 W was applied to the bottom surface of the heat sink base. The ambient environment and fluid inlet temperatures were both set to a reference value of 293.15 K. For the working fluids, the inlet was given a fluid velocity of 2 m/s to represent forced convection. Additionally, the outlet was constrained with a gauge pressure of zero to ensure that there were no undesirable fluid characteristics.

A no-slip boundary condition was utilized to define a fluid velocity of zero at each fin surface for a resistive boundary layer. Additionally, ideal thermal contact was assumed between the heat source and the bottom of the base plate. Material properties were then assigned with the COMSOL library, using presets of copper, aluminum, water, and air. The COMSOL solver calculated the velocity and pressure fields of the designed domains before finding the temperature distribution. The study only resolved if the values converged and the error within the calculations was negligible.

To ensure all the values given by the study were reasonable, an integrated surface value was obtained for the entire heat sink, minus the bottom of it, of the heat rate output. The value calculated was within 0.001 W, ensuring that there was a conservation of energy in the simulation.

Numerical Results

Efficiency was selected as the primary metric for comparing analytical and COMSOL results because it provides a comprehensive measure of performance, directly incorporating the effects of convection coefficient, temperature difference, and heat transfer rate into a single parameter, making it the most effective way to evaluate overall heat sink performance.

0.1 COMSOL results comparing to analytical solution with air as cooling fluid. Using Eq. 13, the total heat transfer area was calculated as:

$$A_{tot} = (0.06)^2 + 24(0.010)(0.040) = 0.0132\,\mathrm{m}^2 \tag{14}$$

Using Eq. 12, the COMSOL-based overall efficiency for aluminum under air cooling was:

$$\eta_{o,Al} = \frac{90}{(68.50)(0.0132)(454.95 - 293.15)} = 0.6152 \tag{15}$$

Using Eq. 12, the COMSOL-based overall efficiency for copper under air cooling was:

$$\eta_{o,Cu} = \frac{90}{(69.20)(0.0132)(437.41 - 293.15)} = 0.6830 \tag{16}$$

The results show that copper achieves a higher efficiency than aluminum under air cooling (0.6830 vs. 0.6152), indicating that material conductivity plays a significant role when convection is relatively weak.

0.2 COMSOL results comparing to analytical solution with water as cooling fluid. Using Eq. 12, the COMSOL-based overall efficiency for aluminum under water cooling was:

$$\eta_{o,Al,water} = \frac{90}{(50200)(0.0132)(295.38 - 293.15)} = 0.0609 \tag{17}$$

Using Eq. 12, the COMSOL-based overall efficiency for copper under water cooling was:

$$\eta_{o,Cu,water} = \frac{90}{(63400)(0.0132)(294.84 - 293.15)} = 0.0635 \tag{18}$$

Under water cooling, both materials exhibit very low efficiencies (0.06), with only a marginal difference between copper and aluminum. This indicates that the high convection coefficient dominates the heat transfer process, minimizing the impact of solid thermal conductivity.

0.3 MATLAB plots comparing to analytical results. The following plot uses MATLAB to compare the overall fin efficiency of aluminum and copper heat sinks under air cooling conditions, comparing analytical solutions and COMSOL simulation results.

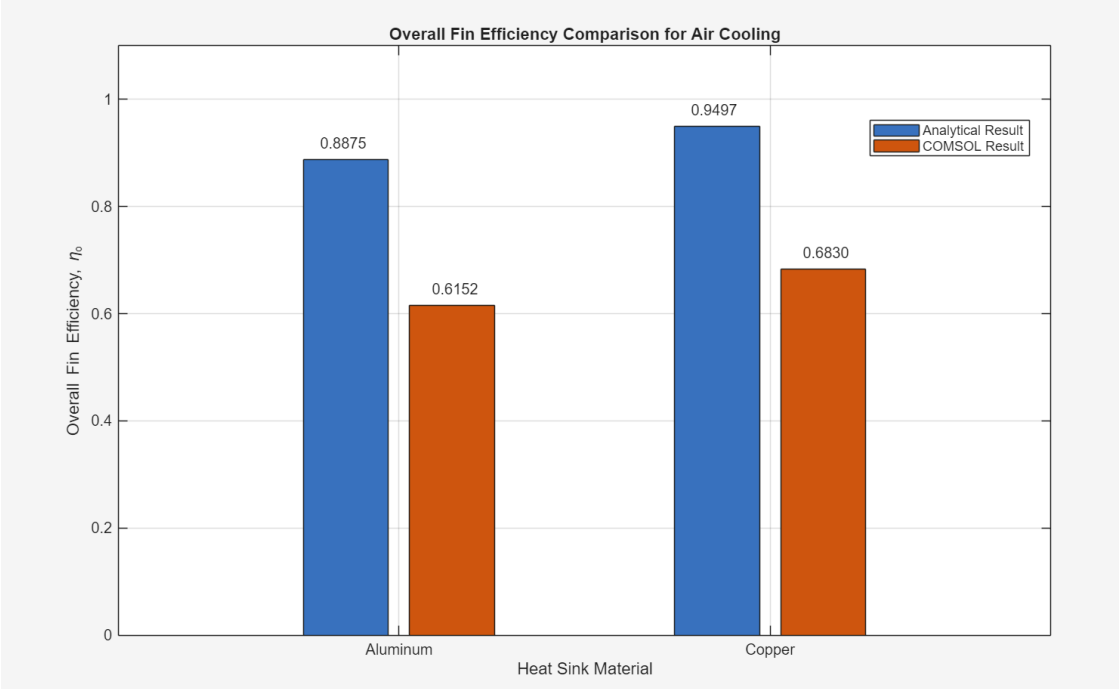


Figure 4 Overall fin efficiency comparison between analytical and COMSOL-based results for aluminum and copper heat sinks under air cooling.

Figure 4 shows that analytical efficiencies are consistently higher than COMSOL results, with copper maintaining higher efficiency than aluminum in both cases. The discrepancy highlights the limitations of the analytical model under realistic flow conditions.

The graph below shows a comparison between the analytical and COMSOL-based overall fin efficiencies for aluminum and copper heat sinks under water cooling conditions.

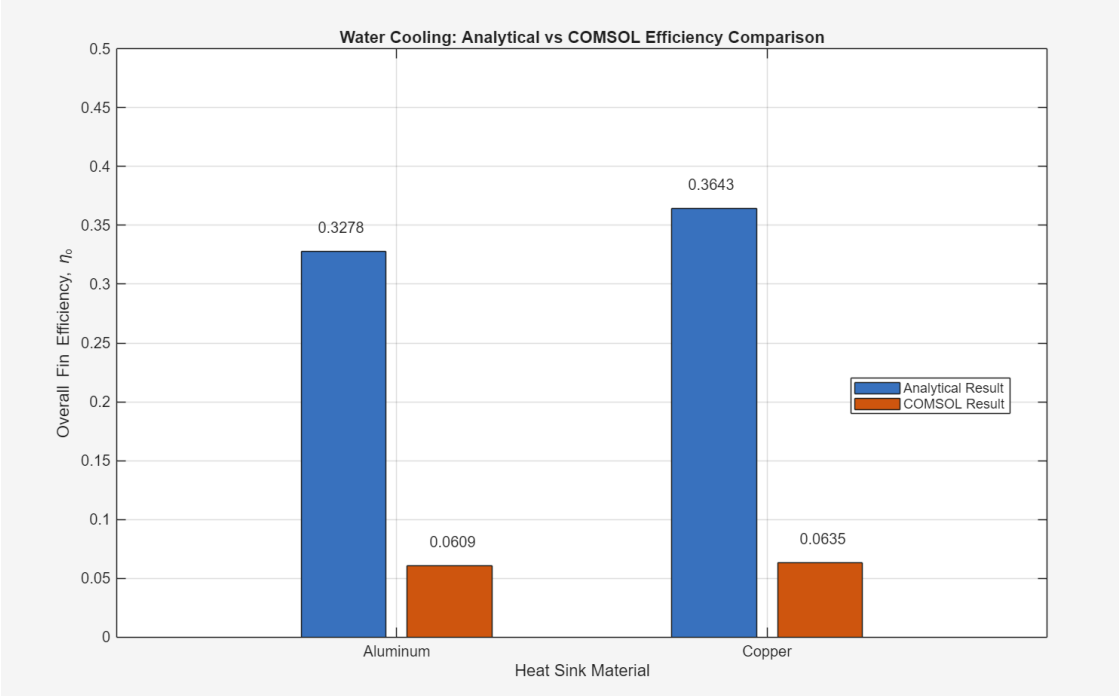


Figure 5 Overall fin efficiency comparison between analytical and COMSOL results for aluminum and copper under water cooling.

Figure 5 shows a significantly larger discrepancy between analytical and COMSOL results for water cooling, with percent errors exceeding 80%. Despite this, both results confirm that copper slightly outperforms aluminum, although the difference is negligible due to the dominance of convective heat transfer.

Comparing air and water cases, the working fluid has a far greater impact on performance than material selection, as evidenced by the drastic reduction in temperature and increase in convection coefficient when using water.

0.4 Percent Error and Analysis. The percent error between the analytical and COMSOL results is calculated using:

$$\% \text{ Error} = \left| \frac{\eta_{\text{analytical}} - \eta_{\text{COMSOL}}}{\eta_{\text{analytical}}} \right| \times 100 \tag{19}$$

For aluminum air cooling:

$$\% \text{ Error}_{\text{Al}} = \left| \frac{0.8875 - 0.6152}{0.8875} \right| \times 100 = 30.68\% \tag{20}$$

For copper air cooling:

$$\% \text{ Error}_{\text{Cu}} = \left| \frac{0.9497 - 0.6830}{0.9497} \right| \times 100 = 28.09\% \tag{21}$$

For aluminum water cooling:

$$\% \text{ Error}_{\text{Al, water}} = \left| \frac{0.3278 - 0.0610}{0.3278} \right| \times 100 = 81.39\% \tag{22}$$

For copper water cooling:

$$\% \text{ Error}_{\text{Cu, water}} = \left| \frac{0.3643 - 0.0636}{0.3643} \right| \times 100 = 82.54\% \tag{23}$$

The COMSOL results for air cooling show lower efficiencies than the analytical predictions, with a percent error of approximately 30% for both materials. This discrepancy can be attributed to several factors. The analytical model assumes idealized one-dimensional heat conduction along the fin, uniform convection coefficients, and simplified boundary conditions such as a perfectly uniform base temperature and ideal tip behavior. In contrast, the COMSOL simulation captures multi-dimensional heat transfer effects, including temperature gradients within both the base plate and fins, which reduces the effective heat transfer performance. Additionally, the analytical solution relies on averaged properties, such as a constant convection coefficient evaluated at a film temperature, whereas COMSOL accounts for spatial variations in temperature and flow. The analytical model also neglects geometric and flow complexities such as fin spacing effects, flow interference between fins, and entrance effects, all of which can reduce heat transfer in practice.

The COMSOL results for water cooling show significantly lower efficiencies than the analytical predictions, with a percent error exceeding 80% for both materials. This larger discrepancy is primarily due to the extremely high convection coefficients and very small temperature differences observed in the COMSOL results. Since efficiency is inversely proportional to $hA\Delta T$, the combination of large h values and small ΔT strongly reduces the calculated efficiency. In the analytical model, the convection coefficient is significantly lower and the temperature difference is larger, leading to a higher predicted efficiency. Similar to the air case, the analytical solution assumes idealized one-dimensional heat transfer, uniform properties, and simplified boundary conditions, while COMSOL captures multi-dimensional effects and realistic gradients. Additionally, the analytical model uses properties evaluated at atmospheric pressure, which affects the estimation of convection coefficients and other fluid properties, whereas COMSOL computes these properties based on the simulation conditions (gauge pressure). Geometric simplifications, such as neglecting fin interaction and flow development between fins, further contribute to overprediction in the analytical model. Finally, the use of maximum temperature as a proxy for base temperature in the COMSOL-based calculation introduces additional error, which is amplified in the water-cooled case due to the small temperature differences involved.

Conclusion

The results demonstrated that a copper heat sink with water as the working fluid performed best. (Although efficiency is lower for water due to its definition, thermal performance is significantly higher). More importantly, the results indicated that the working fluid was the dominant factor in increasing heat sink performance, with water producing temperatures about roughly 150 K lower than

air for both copper and aluminum and average convection coefficients over 700 times larger than air. This is to be expected given that water has a heat transfer coefficient about 10–100 times larger than air. The maximum heat sink temperature of water for both copper and aluminum were about 295 K, indicating that convective heat transfers are so large (63,400 and 50,200, respectively) that the conductive resistance of the heat sink and fins due to the metal material selection is negligible in comparison to the performance and selection of the fluid. For air as the working fluid, copper's higher thermal conductivity had a significant impact, creating a lower maximum temperature by about 17.6 K. When the thermal conductivity of the working fluid is low, then the thermal conductivity of the solid makes a more significant impact to the effectiveness of the design.

The analytical results in comparison show massive discrepancies to the simulated results, which is to be expected. The analytical heat transfer coefficient was under-predicted by an order of magnitude for water and 3 times lower for air. The maximum change in temperature was also predicted to be 321–344 K for air, which is unrealistic. Analytical predictions rely on heavily simplified and ideal flow conditions as well as uniform boundary conditions. The COMSOL model more accurately represents the flow between the fins and captures the developing thermal boundary condition along the plate. Regardless, the analytical model predicted much better performance for water and a higher fin efficiency for copper, which is reflected in the COMSOL results.

Overall, the findings are consistent with theoretical predictions. It was reflected in this study that improving solid conductive properties yielded minimal results when the fluid's convective resistance dominates, which is predicted in theory. The geometry and flow conditions, including the choice of fluid, prove to be the limiting factor for effective heat transfer. Many improvements can be made to both the model as well as the experiment, including varying the geometry of the fins to optimize the flow of the working fluid, testing more materials and working fluids, testing in turbulent flow conditions, and adding a processor with contact resistance. This experiment provides a useful basis for testing and understanding performance of a flat plate and fin heat sink that are often used in electronics. It also emphasizes the necessity of simulation to obtain accurate design information rather than reliance on analytical solutions.

Bibliography

References

[1] Bergman, T.L. and Lavine, A.S., *Fundamentals of Heat and Mass Transfer*, 8th ed., Wiley, 2020.

[2] Sino Extrud, *How Thick Should a Heat Sink Be for Efficient Heat Dissipation?*, [Online]. Available: https://sinoextrud.com/how-thick-should-a-heat-sink-be-for-efficient-heat-dissipation/

[3] Engineering ToolBox, *Convective Heat Transfer*, [Online]. Available: https://www.engineeringtoolbox.com/convective-heat-transfer-d$_4$30.*html*

Cited AI Prompts.

- ChatGPT, response to "Generate MATLAB code to create a bar chart comparing analytical and COMSOL efficiency results for air and water cooling cases," OpenAI, April 2026.
- ChatGPT, response to "Format engineering equations and numerical results into Overleaf-ready LaTeX code with proper referencing," OpenAI, April 2026.
- ChatGPT, response to "Refine technical writing and improve clarity for numerical results, error analysis, and conclusion sections of a heat transfer report," OpenAI, April 2026.

MATLAB Code

Listing 1: MATLAB code used to generate efficiency comparison bar chart for water cooling

```
clc; clear; close all;

% Materials
materials = categorical({'Aluminum','Copper'});
materials = reordercats(materials, {'Aluminum','Copper'});

% Analytical water results (from your table)
eta_analytical_water = [0.3278, 0.3643];

% COMSOL calculated water efficiencies (from earlier)
eta_comsol_water = [0.0609, 0.0635];

% Combine data
eta_data = [eta_analytical_water(:), eta_comsol_water(:)];

% Create bar chart
figure;
bar(materials, eta_data);

% Labels and formatting
ylabel('Overall Fin Efficiency, \eta_o');
xlabel('Heat Sink Material');
title('Water Cooling: Analytical vs COMSOL Efficiency Comparison');

legend('Analytical Result', 'COMSOL Result', 'Location', 'best');

ylim([0 0.5]);
grid on;

% Add value labels
for i = 1:length(materials)
    text(i - 0.15, eta_analytical_water(i) + 0.02, ...
        sprintf('%.4f', eta_analytical_water(i)), ...
        'HorizontalAlignment', 'center', 'FontSize', 10);

    text(i + 0.15, eta_comsol_water(i) + 0.02, ...
        sprintf('%.4f', eta_comsol_water(i)), ...
        'HorizontalAlignment', 'center', 'FontSize', 10);
end
```